From Episodes to Abstractions

Latent Hierarchical Memory in 1,908 AI Conversations

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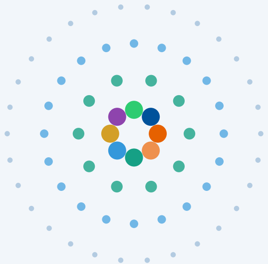

github.com/queelius/chatgpt-complex-net

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Slide 1 – Title (35 s, ∼85w)

Good afternoon. I'm Alex Towell. Joint work with John Matta at SIUE.

In the last three years, millions of people have had thousands of conversations with LLMs. Each one is a trace of an idea being explored. What if we treated these archives not as chat logs, but as externalised semantic memory?

That's the question this paper asks. I'll show that one user's 1,908 ChatGPT conversations contain the same kind of structure cognitive scientists find in human semantic memory.

[Advance.]

The Problem: Archives Are Flat. Knowledge Is Not.

What you have

Chronological list

- Bayesian inference
- Cooking pasta
- Solomonoff induction
- R package debugging
- Philosophy of mind
- ⋮

Structure is invisible.

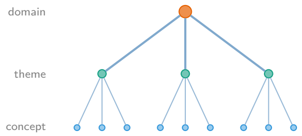
extract → concepts

The Question

Does **structure** emerge from a long archive of AI conversations — the way it does in human memory?

What's hidden

Latent hierarchy



If it's there, we should find it.

Slide 2 – The Problem (50 s, ∼115w)

On the left of the slide, what you actually have when you open your ChatGPT history: a chronological list. Bayesian inference next to cooking pasta next to Solomonoff induction. Adjacency in *time* tells you nothing about which conversations are about the same *thing*.

On the right, what's hidden. You know structure is there — you came back to Solomonoff three times last month; Bayesian inference is part of a domain you've been exploring. That hierarchy is real; it's just invisible from a list.

So: can we recover that structure automatically? And is it the kind of structure cognitive science predicts for semantic memory?

To answer that, I borrow a framework from a different field.

Two Ideas from Cognitive Science

Hippocampal
fast, episodic
specific events

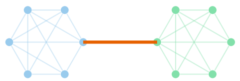
consolidate

Neocortical
slow, statistical
extracts regularities

Prediction:
New conversations reuse *existing* concepts more and more — vocabulary growth **slows**

[McClelland et al. 1995; Kumaran et al. 2016]

And: human semantic memory is a small-world network



Dense local clusters + a few global shortcuts.
Roget's, WordNet, word associations:
small-worldness ≈ 5–15

[Steyvers & Tenenbaum 2005]

The Hypothesis

If AI conversation archives are an *externalised* semantic memory, we should see both signatures: **sublinear vocabulary growth** *and* **small-world topology**.

Slide 3 – Cognitive Framework (1:15, ∼130w)

Two ideas from cognitive science.

On the left, Complementary Learning Systems. The hippocampus encodes individual experiences fast; the neocortex extracts statistical regularities across many of them. As you learn, new experiences increasingly map onto categories you already have. We can measure that saturation as sublinear vocabulary growth.

On the right, the structural side. Steyvers and Tenenbaum mapped human semantic memory as a network and consistently found small-world topology — dense local clusters plus a few long-range shortcuts. Small-worldness scores between roughly 5 and 15.

That gives us the hypothesis at the bottom of the slide: if conversation archives are an externalised semantic memory, we should see *both* signatures. In one user's 1,908-conversation archive.

Here's how we tested it.

Pipeline: Concepts → Embeddings → Hierarchy

- **Raw episodes**: 1,908 conversations, Dec 2022 – Apr 2025 →
- **Extract concepts**: LLM (Claude), 3–10 per episode →
- **Embed**: each concept → 768-d vector →
- **Cluster**: hierarchical, 4-level, episode → . . . → domain

Key instruction in the prompt: *"Be specific. Say 'Metropolis–Hastings algorithm,' not 'statistics.'"*

Output: a graph linking each episode to its concepts, then grouped into a four-level hierarchy.

Slide 4 – Pipeline (1:00, ∼120w)

Following the pipeline from left to right, four stages.

Raw episodes: 1,908 ChatGPT conversations over 29 months from one user — me. Single-user case study; I'll come back to that.

Extract concepts: an LLM pulls three to ten concepts from each episode. The orange box below the pipeline shows the key instruction in the prompt — specificity. Not "statistics"; "Metropolis-Hastings algorithm." Specific concepts are what makes the geometry work later.

Embed: each concept becomes a 768-dimensional vector.

Cluster: hierarchical clustering, cut at three heights.

The line at the bottom names the output: a graph linking episodes to concepts, plus a four-level hierarchy. Everything that follows is just *measuring* this structure.

Here's what came out.

The Reveal: A Four-Level Hierarchy Emerges

1,908 episodes

500 meta-concepts

50 themes

8 domains

Ward linkage on concept embeddings,
cut at 3 heights · ∼3.5× compression per level

Eight interpretable domains

Software Engineering	1,074
Math & Optimisation	634
Statistical Methods	510
Philosophy & AI Theory	489
ML & Network Science	274
DevOps & Integration	273
AI Safety & Research	177
LLM Engineering	86

Two surprises

- Domains are **easy to name** — but there's no clean number of clusters.
- Most episodes **belong to several domains at once**.

Slide 5 – The Reveal (1:00, ∼120w)

[Pause. Let the picture land.]

Reading down the cascade on the left: 1,908 episodes consolidate into 500 meta-concepts, then 50 themes, then 8 domains. Each level groups roughly 3.5 of the level below — the geometric signature of a hierarchy.

The table on the right names the eight domains. I didn't pre-specify them — they emerged from the clustering and reflect my actual research and project history.

Two surprises, in the block below.

First: there's no clean number of clusters. Knowledge is a continuum; eight is a useful summary, not a "right answer."

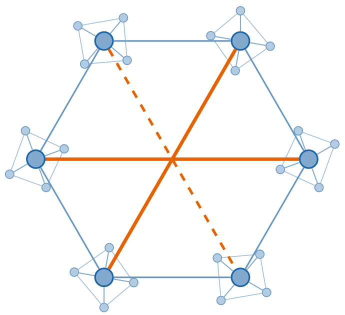
Second: most episodes belong to several domains at once. We'll come back to that.

Cognitive science predicts two signatures here. Both have to land for the analogy to hold. Here's the first.

Signature 1: Small-World Topology

Concept co-occurrence network

499 meta-concept nodes · 5,935 co-occurrence edges



Node: 1 of 500 meta-concepts
Edge: both concepts appear in the same episode

Compared to a random graph

[Humphries & Gurney 2008]

Local clustering	**6.89**×
Shortest paths	1.05×
Small-worldness	**6.6**

Matches human memory

(small-worldness)

Roget's Thesaurus	≈ 13
WordNet	≈ 15
Word associations	≈ 5.6
Our archive	**≈ 6.6**

[comparison values: Steyvers & Tenenbaum 2005]

Slide 6 – Small-World Topology (45 s, ∼95w)

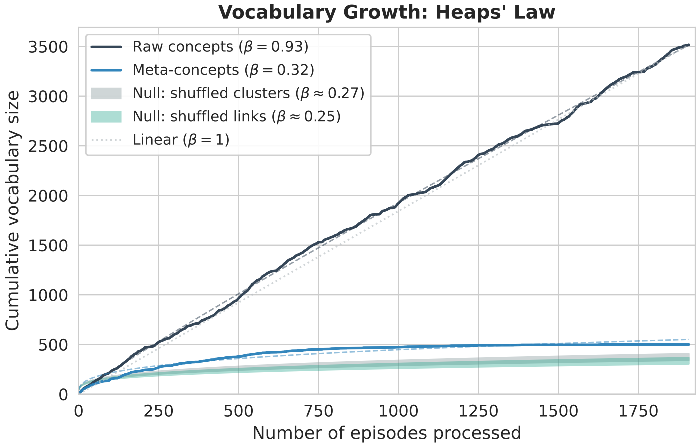
Quick recap: these are the same 500 meta-concepts from the cascade, now drawn as a network. A node is one meta-concept. An edge means the two showed up in the same conversation. So if one chat covered both Bayesian inference and MCMC, those nodes get linked — even across domains.

The cartoon shape — tight local clusters with a few long-range shortcuts crossing the middle — is the classic small-world signature.

Clustering comes in roughly seven times higher than random. Path lengths are unchanged. Small-worldness lands at 6.6. For comparison: human semantic memory networks score between 5 and 15. So we sit right in that range.

I'm not claiming archives *are* semantic memory — just that they show the same structural fingerprint. Signature one, done.

Signature 2: Sublinear Vocabulary Growth



Heaps' law: $V(n) = K \cdot n^{\beta}$

[Heaps 1978]

Raw concepts	$\beta = 0.93$
Meta-concepts	$\boldsymbol{\beta = 0.32}$

Caveat: with *k* fixed at 500

Some saturation is automatic. So we compare against **null models**.

Two nulls, both reject

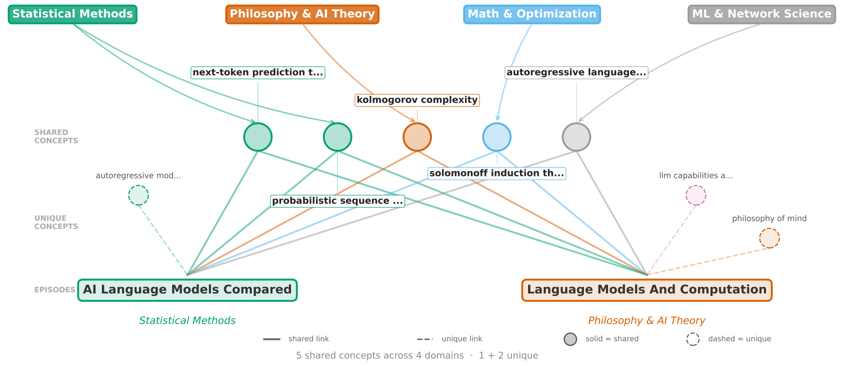
- Shuffled clusters: $\beta = 0.27$
- Shuffled links: $\beta = 0.25$
- Both $p < 10^{-3}$

Slide 7 – Vocabulary saturates (30 s, ∼70w)

This is Heaps' law: vocabulary grows like *n* raised to beta. Real text usually sits at beta around 0.4 to 0.6 — and lower means stronger saturation.

The top curve is raw concepts at beta 0.93 — nearly linear. The bottom curve is meta-concepts at beta 0.32 — strongly sublinear. Two null models both come in below ours, so the saturation is real — not just an artifact of fixing *k* at 500.

77% of Episodes Span Multiple Domains



One-label methods *(Louvain, k-means, BERTopic)*: each episode gets one cluster. These two end up apart — connection vanishes.

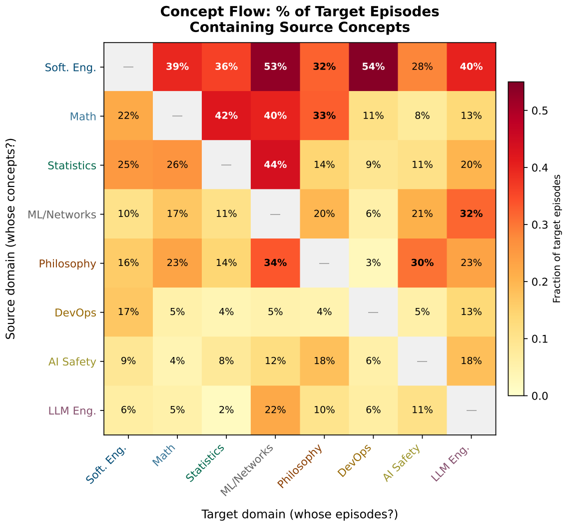
Set-based methods: each episode is a *set* of concepts. Two episodes share concepts across *four* domains and stay linked.

Slide 8 – 77% span multiple domains (40 s, ∼90w)

Look at the two episodes shown in the figure. One came from a conversation about statistical methods; the other from philosophy of AI — so they're in different domains. But they share five concepts in common, and those concepts actually span *four* different domains between them.

If you used any one-label clustering method — Louvain, *k*-means, BERTopic — you'd put these in separate buckets and that link would just disappear. Set-based clustering keeps them connected. And this isn't rare: 77% of episodes span two or more domains.

Who Borrows From Whom? (Concept Flow)



Two questions per domain

Where do its concepts go? (rows)

Which concepts come in? (columns)

What the heatmap says

- On average **39% below** random
  $\Rightarrow$ domains really are separate
- LLM Eng. $\to$ ML/Net: **2.67**$\times$ random
  $\Rightarrow$ specific dependencies

Software Eng. broadcasts to all 7 others.

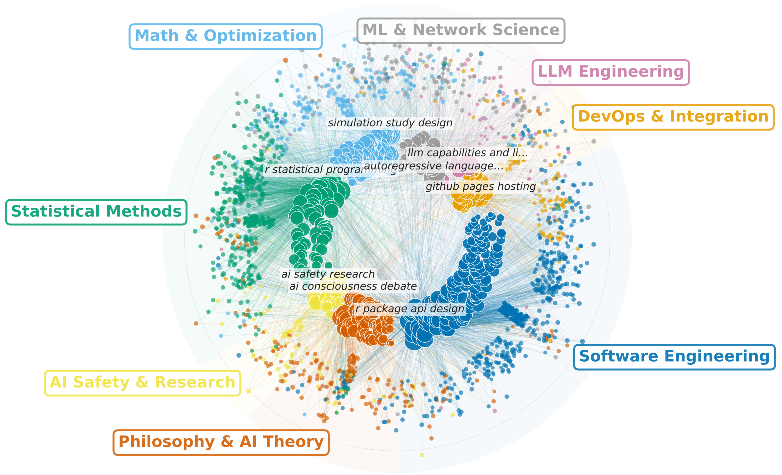
Slide 9 – Who borrows from whom (40 s, ~90w)

The heatmap shows directional concept flow between domains. Reading across a row tells you where that domain's concepts flow out to; reading down a column tells you what comes in. That asymmetry is something an undirected network can't capture.

After controlling against a size-preserving null, sharing actually comes in **39% below random** — so domains really are separate.

But one outlier survives that control: LLM Engineering feeds ML and Networks at **2.67 times** the random expectation. That's a real targeted dependency, not spillover.

The Full Picture



Inner: 499 concepts (coloured by domain). Outer: 1,908 episodes. Edges: episode→concept.

Slide 10 – The Full Picture (20 s, ∼40w)

This is the whole archive in one image: 499 concepts in the inner ring, all 1,908 episodes around the outside, coloured by domain. The point isn't to read individual lines — it's to see that this is structured, not random.

[Brief pause.]

Takeaways & What's Next

What we found

- 4-level hierarchy
  1,908 $\rightarrow$ 500 $\rightarrow$ 50 $\rightarrow$ 8
- Small-world structure matches human memory
- Vocabulary saturates as CLS predicts
- 77% span $\geq$ 2 domains

Why it matters

- Archives = **externalised semantic memory**
- **Concept-level retrieval** (Graph RAG)
- Knowledge is a **continuum, not a partition**

Limits & next

- **N=1** single user. Replicate on WildChat
- **Cluster count** is a design choice

Treat AI conversation archives as **structured cognitive artifacts** —
the structure was there all along; we just needed the right lens.

Thank you

Slide 11 – Takeaways (45 s, ∼95w)

To wrap up, three findings. A four-level hierarchy emerged from this one user's archive. Small-world topology and saturating vocabulary — the same signatures cognitive science predicts for human semantic memory. And 77% of episodes span multiple domains, which is why set-based clustering captures what one-label methods would lose.

The headline is at the bottom of the slide: AI conversation archives are structured cognitive artifacts. The structure was there all along; we just needed the right lens.

Thank you for watching.